JavaScript is disabled on your browser.

compact1, compact2, compact3

java.util

Class OptionalDouble

java.lang.Object
    java.util.OptionalDouble

```
public final class OptionalDouble
extends Object
```

A container object which may or may not contain a `double` value. If a value is present, `isPresent()` will return `true` and `getAsDouble()` will return the value.

Additional methods that depend on the presence or absence of a contained value are provided, such as `orElse()` (return a default value if value not present) and `ifPresent()` (execute a block of code if the value is present).

This is a value-based class; use of identity-sensitive operations (including reference equality (==), identity hash code, or synchronization) on instances of `OptionalDouble` may have unpredictable results and should be avoided.

Since:

1.8

Method Summary

Concrete Methods

Modifier and Type	Method	Description
static `OptionalDouble`	`empty()`	Returns an empty `OptionalDouble`
boolean	`equals(Object obj)`	

		Indicates whethe other object is "e this OptionalDou
double	**getAsDouble**()	If a value is presen OptionalDouble, the value, otherw throws NoSuchElementE
int	**hashCode**()	Returns the hash value of the pres if any, or 0 (zero) value is present.
void	**ifPresent**(**DoubleConsumer** consumer)	Have the specifie consumer accept value if a value is otherwise do not
boolean	**isPresent**()	Return true if th value present, ot false.
static **OptionalDouble**	**of**(double value)	Return an OptionalDouble specified value p
double	**orElse**(double other)	Return the value present, otherwis other.
double	**orElseGet**(**DoubleSupplier** other)	Return the value present, otherwis other and return result of that inv
<X extends **Throwable**> double	**orElseThrow**(**Supplier**<X> exceptionSupplier)	Return the conta value, if present, otherwise throw exception to be c the provided sup
String	**toString**()	Returns a string representation o object.

Methods inherited from class java.lang.Object

clone, finalize, getClass, notify, notifyAll, wait, wait, wait

Method Detail

empty

```
public static OptionalDouble empty()
```

Returns an empty `OptionalDouble` instance. No value is present for this OptionalDouble.

API Note:

Though it may be tempting to do so, avoid testing if an object is empty by comparing with == against instances returned by Option.empty(). There is no guarantee that it is a singleton. Instead, use `isPresent()`.

Returns:

an empty OptionalDouble.

of

```
public static OptionalDouble of(double value)
```

Return an `OptionalDouble` with the specified value present.

Parameters:

value - the value to be present

Returns:

an OptionalDouble with the value present

getAsDouble

```
public double getAsDouble()
```

If a value is present in this `OptionalDouble`, returns the value, otherwise throws `NoSuchElementException`.

Returns:

the value held by this OptionalDouble

Throws:

NoSuchElementException - if there is no value present

See Also:

`isPresent()`

isPresent

```
public boolean isPresent()
```

Return `true` if there is a value present, otherwise `false`.

Returns:

`true if there is a value present, otherwise false`

ifPresent

```
public void ifPresent(DoubleConsumer consumer)
```

Have the specified consumer accept the value if a value is present, otherwise do nothing.

Parameters:

`consumer - block to be executed if a value is present`

Throws:

`NullPointerException - if value is present and consumer is null`

orElse

```
public double orElse(double other)
```

Return the value if present, otherwise return `other`.

Parameters:

`other - the value to be returned if there is no value present`

Returns:

`the value, if present, otherwise other`

orElseGet

```
public double orElseGet(DoubleSupplier other)
```

Return the value if present, otherwise invoke `other` and return the result of that invocation.

Parameters:

other - a DoubleSupplier whose result is returned if no value is present

Returns:

the value if present otherwise the result of other.getAsDouble()

Throws:

NullPointerException - if value is not present and other is null

orElseThrow

```
public <X extends Throwable> double orElseThrow(Supplier<X> exceptionSupplier)
                                                throws X
```

Return the contained value, if present, otherwise throw an exception to be created by the provided supplier.

API Note:

A method reference to the exception constructor with an empty argument list can be used as the supplier. For example, IllegalStateException::new

Type Parameters:

X - Type of the exception to be thrown

Parameters:

exceptionSupplier - The supplier which will return the exception to be thrown

Returns:

the present value

Throws:

X - if there is no value present

NullPointerException - if no value is present and exceptionSupplier is null

X extends Throwable

equals

```
public boolean equals(Object obj)
```

Indicates whether some other object is "equal to" this OptionalDouble. The other object is considered equal if:

- it is also an OptionalDouble and;
- both instances have no value present or;
- the present values are "equal to" each other via Double.compare() == 0.

Overrides:

`equals` in class `Object`

Parameters:

`obj - an object to be tested for equality`

Returns:

`{code true} if the other object is "equal to" this object otherwise false`

See Also:

`Object.hashCode()`, `HashMap`

hashCode

```
public int hashCode()
```

Returns the hash code value of the present value, if any, or 0 (zero) if no value is present.

Overrides:

`hashCode` in class `Object`

Returns:

`hash code value of the present value or 0 if no value is present`

See Also:

`Object.equals(java.lang.Object)`, `System.identityHashCode(java.lang.Object)`

toString

```
public String toString()
```

Returns a string representation of the object. In general, the `toString` method returns a string that "textually represents" this object. The result should be a concise but informative representation that is easy for a person to read. It is recommended that all subclasses override this method.

The `toString` method for class `Object` returns a string consisting of the name of the class of which the object is an instance, the at-sign character `@', and the unsigned hexadecimal representation of the hash code of the object. In other words, this method returns a string equal to the value of:

```
getClass().getName() + '@' + Integer.toHexString(hashCode())
```

Returns a non-empty string representation of this object suitable for debugging. The exact presentation format is unspecified and may vary between implementations and versions.

Overrides:

`toString` in class `Object`

Implementation Requirements:

If a value is present the result must include its string representation in the result. Empty and present instances must be unambiguously differentiable.

Returns:

the string representation of this instance

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